



BAULKHAM HILLS HIGH SCHOOL

2013

HSC Assessment Task 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

Total marks –

This paper consists of TWO sections.

Section 1 – Multiple Choice

4 Marks: Answer by shading the appropriate circle in your answer booklet

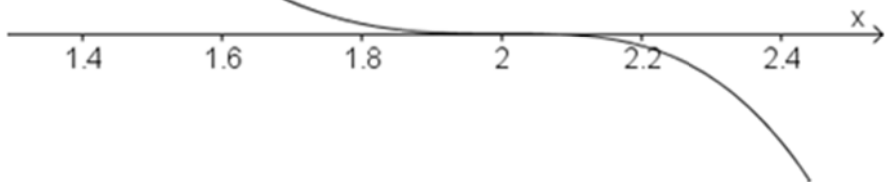
Section 2 – Extended Response

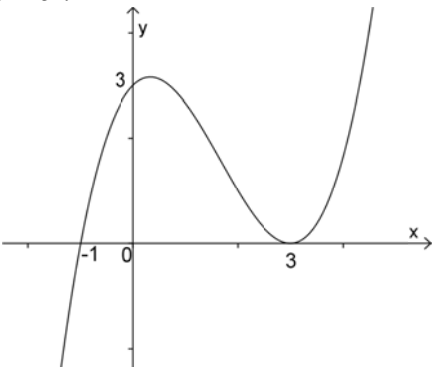
33 marks

Attempt all questions

Answer each question on the appropriate page of your answer booklet. Show all necessary working.

Section 1 – Multiple choice (4 marks)**Attempt all questions.**

		Marks
1	If two polynomials have degrees m and n respectively, and $m < n$, what is the maximum number of points of intersection of their graphs? (A) m (B) n (C) $m+n$ (D) $n-m$	1
2	The polynomial $P(x) = 2x^3 - 9x^2 + 13x + k$ is divisible by $x - 2$, and k is a constant. $P(x)$ is also divisible by: (A) $2x - 1$ (B) $x + 1$ (C) $x - 1$ (D) $x + 2$	1
3	If one root of the equation $x^3 - 5x^2 + 5x - 1 = 0$ is $2 - \sqrt{3}$, then the sum of the other two roots is: (A) $-7 + \sqrt{3}$ (B) $-1 + \sqrt{3}$ (C) $3 + \sqrt{3}$ (D) $-3 + \sqrt{3}$	1
4	Part of the graph $y = P(x)$, where $P(x)$ is a polynomial of degree four is shown below.  Which of the following could be the polynomial $P(x)$? (A) $P(x) = x(x - 2)^3$ (B) $P(x) = (x - 2)^3(x + 3)$ (C) $P(x) = (2 - x)^3(3 - x)$ (D) $P(x) = (x - 1)(x - 2)^3$	1
End of Section 1		
Section 2 – Extended Response Attempt all questions on the appropriate page of your answer booklet. Show all necessary working.		
5	When the polynomial $P(x) = x^3 + ax^2 - 4x + 1$ (where a is a constant) is divided by $x - 1$, the remainder is 2. What is the remainder with $P(x)$ is divided by $2x - 1$?	2

6	Prove by Mathematical Induction that $3^{2n+4} - 2^{2n}$ is divisible by 5 for all positive integers n	3
7	The graph below shows all the important features of a polynomial. Find an equation for this polynomial. 	3
8	α, β and γ are the roots of $2x^3 - 6x^2 + x - 9 = 0$. Evaluate: (i) $\alpha + \beta + \gamma$ (ii) $\alpha\beta + \beta\gamma + \gamma\alpha$ (iii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ (iv) $\alpha^2 + \beta^2 + \gamma^2$	1 1 2 2
9	Given $P(x) = 2x^3 + 7x^2 - 46x + 21$: (i) Show that $x - 3$ is a factor of $P(x)$ (ii) Fully factorise $P(x)$ (iii) Hence or otherwise, solve: $2x^3 + 7x^2 - 46x + 21 \geq 0$ (iv) Solve the inequality: $\frac{2x^3 + 7x^2 - 46x + 21}{x} > 0$	1 2 2 2
10	Prove by Mathematical Induction that $\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$ for all positive integers n.	4
11	For $P(x) = x^3 + ax^2 + 2ax + b$, where a and b are constants, the roots are $x = 2, -3, \gamma$. Find the value of a, b and γ .	4
12	If $P(x) = x^4 - 6x^3 + x^2 - 4x - 4$ and $A(x) = x^2 + x + 1$: (i) Find the quotient and remainder when $P(x)$ is divided by $A(x)$ (ii) Hence find two possible polynomials $L(x)$ such that $x^2 + x + 1$ is a factor of $P(x) + L(x)$	2 2
End of Task		

HSC Task 1 : Ext. 1

~ Q5 & 6 - page 1 ~

Question 5

2013

BOS#: SOLUTIONS.

$$P(1) = 1 + a - 4 + 1 = 2$$

$$a - 2 = 2$$

$$a = 4$$

$$P(x) = x^3 + 4x^2 - 4x + 1$$

$$\text{Remainder} = P\left(\frac{1}{2}\right) = \frac{1}{8} + 1 - 2 + 1$$

$$= \frac{1}{8}$$

Question 6

$$\text{If } n=1 \quad 3^{2n+4} - 2^{2n} = 3^6 - 2^2$$

$$= 729 - 4$$

$$= 725 \quad \text{which is divisible by 5}$$

$\therefore$ True for $n=1$

Assume true for $n=k$.

ie Assume : $3^{2k+4} - 2^{2k} = 5m$ for some integer m .

Now need to prove true for $n=k+1$:

$$3^{2(k+1)+4} - 2^{2(k+1)} = 3^{2k+6} - 2^{2k+2}$$

$$= 3^2 \cdot 3^{2k+4} - 2^2 \cdot 2^{2k}$$

$$= 9(3^{2k+4}) - 4(2^{2k})$$

$$= 9(3^{2k+4} - 2^{2k}) + 5(2^{2k}) \quad \text{by assumption}$$

$$= 9(5m) + 5(2^{2k})$$

$$= 5(9m + 2^{2k})$$

$$= 5p \quad \text{where } p = 9m + 2^{2k} \text{ is an integer.}$$

If true for $n=k$, then also true for $n=k+1$
 Statement is true for $n=1$

$\therefore$ Also true for $n=2, 3, 4 \dots$ and by induction,
 true for all positive integers n .

($n=1$, assump, cond.) ——— 1

Mult. Choice : Working

1. The larger degree $= n \therefore$ (B)

2. $P(2) = 16 - 36 + 26 + k = 0 \therefore k = -6$

$$P(x) = 2x^3 - 9x^2 + 13x - 6$$

$$P\left(\frac{1}{2}\right) \neq 0$$

$$P(-1) \neq 0$$

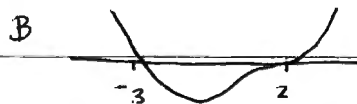
$$P(1) = 0 \therefore x-1 \text{ is a factor}$$

(C)

3. $\alpha + \beta + 2 - \sqrt{3} = 5$

$$\alpha + \beta = 5 - 2 + \sqrt{3} = 3 + \sqrt{3} \quad (C)$$

4.



(C)

You may ask for extra writing paper if you need more space to answer question 5 & 6

Question 7

BOS#:

$$y = a(x+1)(x-3)^2$$

— 1 for (x+1)

— 1 for (x-3)²

Sub. (0, 3) :

$$3 = a \times 1 \times (-3)^2$$

$$3 = 9a$$

$$a = \frac{1}{3}$$

— 1

$$\therefore y = \frac{1}{3}(x+1)(x-3)^2$$

Question 8

$$2x^3 - 6x^2 + x - 9 = 0$$

$$(i) \alpha + \beta + \gamma = \frac{-b}{a} = \frac{6}{2} = 3$$

1

$$(ii) \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = \frac{1}{2}$$

1

$$(iii) \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

1

$$= \frac{\frac{1}{2}}{\frac{9}{2}}$$

$$= \frac{1}{9}$$

1

$$(iv) \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = 3^2 - 2\left(\frac{1}{2}\right) = 8$$

You may ask for extra writing paper if you need more space to answer question 7 & 8

1

1

Question 9

BOS#:

$$P(x) = 2x^3 + 7x^2 - 46x + 21$$

$$(i) \quad P(3) = 2(3)^3 + 7(3)^2 - 46(3) + 21$$

$$= 54 + 63 - 138 + 21$$

$$= 0 \quad \therefore \quad x - 3 \quad \text{is a factor}$$

$$(ii) \quad \begin{array}{r} 2x^2 + 13x - 7 \\ x-3 \overline{) 2x^3 + 7x^2 - 46x + 21} \end{array} \quad (\text{or alternate method})$$

$$2x^3 - 6x^2$$

$$13x^2 - 46x$$

$$13x^2 - 39x$$

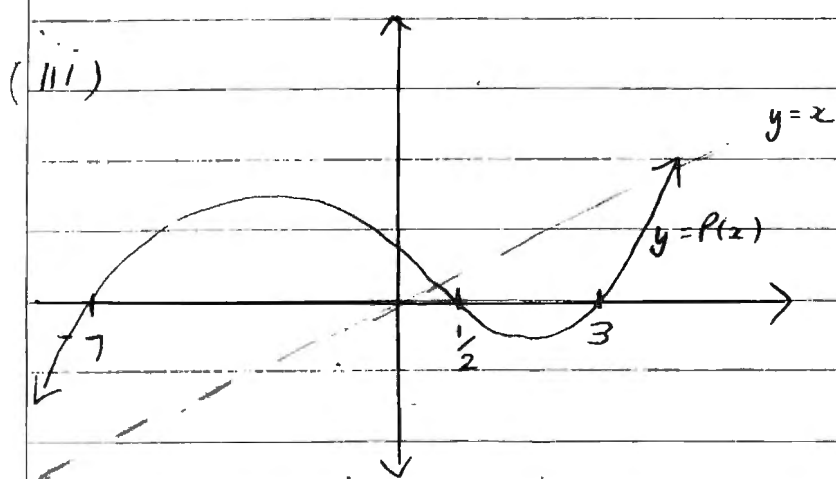
$$-7x + 21$$

$$-7x + 21$$

$$0$$

$$\therefore P(x) = (x-3)(2x^2 + 13x - 7)$$

$$= (x-3)(2x-1)(x+7)$$



$P(x) \geq 0$ when:

$$-7 \leq x \leq \frac{1}{2}, \quad x \geq 3$$

(iv) Fraction > 0 when numerator + denom. have the same sign.

$$0 < x < \frac{1}{2}, \quad x > 3$$

$$x < -7$$

both pos

both neg

~~NA~~

~~(2x fully correct)~~

Question 10

BOS#:

$$\text{If } n=1, \text{ LHS} = \frac{1}{1 \cdot 2 \cdot 3} = \frac{1}{6}$$

LHS = RHS

$$\text{RHS} = \frac{1 \times 4}{4 \times 2 \times 3} = \frac{4}{24} = \frac{1}{6}$$

$\therefore$ True for $n=1$

Assume true for $n=k$:

ie. Assume

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots + \frac{1}{k(k+1)(k+2)} = \frac{k(k+3)}{4(k+1)(k+2)}$$

Prove true for $n=k+1$

ie. Prove

$$\begin{aligned} & \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \\ &= \frac{(k+1)(k+4)}{4(k+2)(k+3)} \end{aligned}$$

$$\text{LHS} = \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \quad \text{by assumption}$$

$$= \frac{k(k+3)^2 + 4}{4(k+1)(k+2)(k+3)}$$

$$= \frac{k^3 + 6k^2 + 9k + 4}{4(k+1)(k+2)(k+3)}$$

Now numerator has a factor of $x+1$, since $(-1)^3 + 6(-1)^2 + 9(-1) + 4 = -1 + 6 - 9 + 4 = 0$

$$\text{Numerator} = (k+1)(k^2 + mk + 4)$$

and since $mk + 4k = 9k$, $m = 5$

$$\text{Numerator} = (k+1)(k^2 + 5k + 4)$$

$$= (k+1)(k+1)(k+4)$$

OR EXPANSION

OR

LONG

DIVISION.

$$\text{LHS} = \frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)} = \text{RHS}$$

[Justification
of $(k+1)^2(k+4)$
needed]

$\therefore$ If true for $n=k$, then also true for $n=k$

Etc

(No mark for conclusion)

You may ask for extra writing paper if you need more space to answer question 10

Question 11

BOS#:

$$2 - 3 + \gamma = -a$$

$$\gamma = 1 - a \quad \text{--- (1)}$$

$$-6 - 3\gamma + 2\gamma = 2a$$

$$-\gamma = 2a + 6$$

$$\gamma = -2a - 6 \quad \text{--- (2)}$$

$$-6\gamma = -b$$

$$b = 6\gamma \quad \text{--- (3)}$$

From (1), (2)

$$1 - a = -2a - 6$$

(Any 2 eqns)

$$\underline{a = -7} \quad \text{--- 1}$$

Into (1)

$$\gamma = 1 - (-7) = \underline{8} \quad \text{--- 1}$$

Into (3)

$$b = 6(8) = \underline{48} \quad \text{--- 1}$$

You may ask for extra writing paper if you need more space to answer question 11

Question 12

BOS#:

(i)

$$\begin{array}{r}
 x^2 - 7x + 7 \quad \leftarrow \text{quotient} \quad | \\
 x^2 + x + 1 \overline{) x^4 - 6x^3 + x^2 - 4x - 4} \\
 \underline{x^4 + x^3 + x^2} \\
 -7x^3 - 4x \\
 \underline{-7x^3 - 7x^2 - 7x} \\
 7x^2 + 3x - 4 \\
 \underline{7x^2 + 7x + 7} \\
 -4x - 11 \quad \leftarrow \text{remainder} \quad |
 \end{array}$$

(ii) Any two expressions equivalent to $k(x^2 + x + 1) + 4x + 11$ ($k = \text{integer}$)

Expected answers:

$(k=0)$	$4x + 11$	} 2 (any two)
$(k=1)$	$x^2 + 5x + 12$	
$(k=2)$	$2x^2 + 6x + 13$	

or $S(x)(x^2 + x + 1) + 4x + 11$ where $S(x)$ is a polynomial.

You may ask for extra writing paper if you need more space to answer question 12